

ALADIN AI PRODUCT ROADMAP (6 MONTHS)

Voice AI Commerce + Vietnamese OCR + AI Agent Automation

GO LIVE: Month 6 (April 2026)

EXECUTIVE SUMMARY

Mission: Launch AI-powered mobile app in 6 months that transforms ordering from 10 minutes → 10 seconds

Timeline: November 2025 → April 2026 (GO LIVE)

Investment: \$120-150K

Team: Scale from 3 → 8 people

Target: 500 shops using AI app by launch, 70% order via app vs phone

PRODUCT VISION

"Order in 10 seconds, not 10 minutes"

Current Pain (Without AI):

- ❌ Shop owners spend 2 hours/day calling suppliers
- ❌ Manually write orders, check prices, confirm quantities
- ❌ Miss reorders → Stockouts → Lost revenue
- ❌ No insights into what sells best

Future State (With Aladin AI):

-  Speak order in Vietnamese → Done in 10 seconds
 -  Photo old invoice → AI reorders everything automatically
 -  AI predicts stockouts → Suggests reorder proactively
 -  Never miss an order, always have stock
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6-MONTH AGGRESSIVE TIMELINE

MONTH 1 (November): Foundation

Ship: Basic Zalo chatbot (menu-based ordering)

MONTH 2 (December): Voice

Ship: Voice ordering (Vietnamese speech-to-text)

MONTH 3 (January): OCR

Ship: Photo-to-order (Vietnamese invoice OCR)

MONTH 4 (February): AI Intelligence

Ship: Demand forecasting + Recommendations

MONTH 5 (March): Automation

Ship: Auto-reorder + Smart notifications

MONTH 6 (April): GO LIVE

Ship: Full public launch + Mobile app (iOS/Android)

DETAILED MONTH-BY-MONTH PLAN

MONTH 1 (NOVEMBER 2025): FOUNDATION

Theme: "Get Users on Zalo"

Goal

Shift 40% of orders from phone → Zalo chatbot

What to Build

Feature 1: Zalo Chatbot (Menu-Based)

User flow:

1. Shop owner opens Zalo, finds Aladin OA
2. Sends "Đặt hàng" (Order)
3. Bot shows menu:
 - 1 Gạo (Rice)
 - 2 Dầu ăn (Cooking oil)
 - 3 Mì ăn liền (Instant noodles)
 - 4 Xem thêm (More)
4. User types "1"
5. Bot shows rice options: A) ST25 B) Jasmine C) Nếp
6. User picks, enters quantity
7. Bot confirms: "Tổng: 2.5M VND. Xác nhận?"
8. Order placed 

Ordering time: 60 seconds (vs 5 minutes phone call)

Feature 2: Order History

- View past 10 orders
- 1-tap reorder from history
- Filter by product, date

Feature 3: Order Tracking

- Real-time status: Processing → Packed → Out for Delivery → Delivered
- Delivery ETA (based on driver GPS)
- Zalo notification at each stage

Feature 4: Product Catalog

- Browse all products by category

- Search by product name
- View prices, stock availability

Tech Stack

- **Platform:** Zalo Official Account (OA) + Zalo Mini App
- **Backend:** Node.js + Express
- **Database:** PostgreSQL (customer data, orders, products)
- **Hosting:** AWS (EC2 + RDS)
- **No AI yet** - Pure if-then logic

Team (Month 1)

- Karthick (CTO): Architecture, oversee development
- 1 Mobile Dev: Zalo Mini App integration
- 1 Backend Dev: API development
- 1 Part-time Designer: UI/UX for Zalo interface

Total: 3-4 people

Success Metrics

-  200 customers onboarded to Zalo OA
-  40% of orders via Zalo (vs 0% today)
-  Average order time: <90 seconds
-  85%+ customer satisfaction with chatbot

Budget: \$15-20K

- Team: \$12K (salaries)
 - Infrastructure: \$2K (AWS, Zalo OA fees)
 - Testing: \$1K (user testing incentives)
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MONTH 2 (DECEMBER 2025): VOICE INPUT

Theme: "Speak, Don't Type"

Goal

20% of orders via voice (for shop owners who can't type fast)

What to Build

Feature 5: Voice Ordering

User flow:

1. Shop owner clicks mic button in Zalo
2. Speaks: "Tôi muốn đặt 10 bao gạo ST25" (I want to order 10 bags ST25 rice)
3. AI converts speech → text (Vietnamese)
4. Bot extracts: Product = "gạo ST25", Quantity = 10
5. Bot confirms: "10 bao gạo ST25, đúng không?" (Correct?)
6. User: "Đúng rồi" (Yes)
7. Order placed 

Ordering time: 15 seconds (6x faster than typing!)

Feature 6: Voice Confirmation

- Bot sends voice message confirming order
- Shop owner can reply by voice (no typing)

Feature 7: Multi-Product Voice Ordering

User: "Tôi cần 10 gạo, 5 dầu, 20 mì" (I need 10 rice, 5 oil, 20 noodles)

Bot: "10 gạo ST25, 5 dầu ăn Simply, 20 mì Hảo Hảo, đúng không?"

User: "Đúng"

Feature 8: Vietnamese Dialect Support

- Southern dialect (Saigon, Mekong Delta) - Priority
- Central/Northern dialect support (fallback to standard Vietnamese)

Tech Stack

- **Speech-to-Text:**
 - Primary: Zalo Voice API (built-in, free)
 - Backup: Google Cloud Speech-to-Text (Vietnamese)
- **NLP:** spaCy Vietnamese (intent + entity extraction)
- **Text-to-Speech:** Zalo TTS (for voice confirmations)

AI/ML Work (Simple)

- Intent classification: "Order" vs "Track" vs "Ask question"
- Entity extraction: Product name, quantity, brand
- Training data: Manually label 500 voice messages
- Accuracy target: 80% (good enough for beta)

Team (Month 2)

- Add: 1 AI/ML Engineer (part-time contractor)
- Existing: Karthick + 2 devs + designer

Total: 4-5 people

Success Metrics

-  20% of orders via voice (50+ customers)
-  80% voice recognition accuracy
-  Customer feedback: "Easier than typing"
-  50% of orders via Zalo (voice + chat combined)

Budget: \$20-25K

- Team: \$18K (add ML engineer)
 - APIs: \$3K (Google Cloud, if needed)
 - Testing: \$2K (voice data collection, labeling)
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MONTH 3 (JANUARY 2026): PHOTO ORDERING (OCR)

Theme: "Snap & Order"

Goal

15% of orders via photo (easiest method for repeat orders)

What to Build

Feature 9: Invoice Photo Upload

User flow:

1. Shop owner opens Aladin app in Zalo
2. Clicks "Đặt hàng bằng ảnh" (Order by photo)
3. Uploads photo of old supplier invoice
4. AI processes image (2-3 seconds)
5. Bot shows extracted items:
 - Gạo ST25, 10 bao, 2.5M VND
 - Dầu Simply, 5 chai, 750K VND
6. Bot: "Đặt lại những sản phẩm này?" (Reorder these?)
7. User: "Có" (Yes)
8. Order placed 

Ordering time: 10 seconds (fastest method!)

Feature 10: OCR Product Matching

- AI extracts text from invoice
- Fuzzy matching: "Gao ST25" → "Gạo ST25 (10kg)"
- Handle typos, abbreviations, brand variations
- Confidence score: Show matches AI is unsure about

Feature 11: Manual Correction

- If OCR misreads, user can tap to correct
- User feedback improves model over time

Feature 12: Multi-Invoice Support

- Upload multiple invoice photos at once
- AI combines all items into one order

Tech Stack

- **OCR Engine:**
 - Primary: PaddleOCR (Vietnamese-optimized, open source)
 - Backup: Google Cloud Vision API (higher accuracy, paid)
- **Image Processing:** OpenCV (pre-processing for better OCR)
- **Product Matching:** FuzzyWuzzy + Custom rules (brand aliases)

AI/ML Work

- OCR model: Fine-tune PaddleOCR on Vietnamese invoices
- Training data: Collect 300 real invoices from customers
- Product matching: Build lookup table (aliases, misspellings)
- Accuracy target: 85% (human-in-loop for 15%)

Team (Month 3)

- ML Engineer now full-time (OCR focus)
- Existing: Karthick + 2 devs + designer

Total: 5 people

Success Metrics

-  85% OCR accuracy (product name + quantity)
-  15% of orders via photo
-  <3 seconds processing time per invoice
-  60% of orders via Zalo (voice + photo + chat)

Budget: \$25-30K

- Team: \$20K (ML engineer full-time)
 - Infrastructure: \$3K (GPU servers for OCR)
 - APIs: \$2K (Google Vision for complex invoices)
 - Data collection: \$2K (buy invoice samples)
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MONTH 4 (FEBRUARY 2026): AI INTELLIGENCE

Theme: "Predict & Recommend"

Goal

AI proactively suggests orders before shop runs out

What to Build

Feature 13: Demand Forecasting

How it works:

1. AI analyzes shop's order history (3+ months)
2. Detects patterns:
 - "Shop X orders rice every 12 days"
 - "Orders increase 20% on Fridays"
3. Predicts: "Your rice will run out in 3 days"
4. Sends notification: "Đặt gạo bây giờ để không hết hàng" (Order rice now to avoid stockout)
5. 1-tap to order 

Impact: Never run out of stock again

Feature 14: Smart Reorder Suggestions

- "Based on your sales, you'll need rice by Thursday"
- "You usually order 10 bags, but demand is up. Try 12?"
- Learns from acceptance rate (if user always ignores, stop suggesting)

Feature 15: Personalized Recommendations

Examples:

- "Shops like yours also buy [Product X]"
- "Try snacks - your neighbors sell 30% more with them"
- "Mooncakes coming - order now for Mid-Autumn festival"

Goal: Increase AOV by 15-20%

Feature 16: New Product Alerts

- When Aladin adds new products, notify relevant shops
- "We now have [Brand X] - want to try?"

Tech Stack

- **Forecasting:** Prophet (Facebook's time-series library)
- **Recommendations:**
 - Collaborative filtering (shops similar to you)
 - Content-based (product categories you buy)
- **Backend:** Python (ML models) + FastAPI (serve predictions)

AI/ML Work

Model 1: Demand Forecasting

- Input: Order history (product, quantity, date)
- Output: Predicted stockout date
- Algorithm: Prophet (handles seasonality, holidays)
- Training: Per customer, per product (3+ months data)
- Retraining: Weekly

Model 2: Recommendation Engine

- Input: Customer profile, order history, similar customers
- Output: Top 5 recommended products
- Algorithm: Matrix factorization + content-based
- Training: All customers (collaborative filtering)
- Retraining: Daily

Team (Month 4)

- ML Engineer focuses on forecasting + recommendations
- Backend dev builds ML serving infrastructure
- Existing team continues app features

Total: 5 people

Success Metrics

-  70% forecast accuracy (predicted stockout $\pm$ 1 day)
-  30% customers act on AI suggestions
-  AOV increase: +15% (from recommendations)
-  40% of ordering decisions AI-driven

Budget: \$30-35K

- Team: \$25K
- Infrastructure: \$3K (ML servers)
- Training: \$2K (data science tools, MLflow)

MONTH 5 (MARCH 2026): AUTOMATION

Theme: "Set It, Forget It"

Goal

30% of shops on auto-reorder (zero manual effort)

What to Build

Feature 17: Auto-Reorder System

User flow:

1. Shop owner enables: "Tự động đặt gạo mỗi 2 tuần" (Auto-order rice every 2 weeks)
2. AI monitors inventory based on past sales
3. 2 days before predicted stockout, AI generates order
4. Sends confirmation: "Chúng tôi sẽ giao gạo thứ 5, ok không?" (We'll deliver rice Thursday, ok?)
5. Shop owner can:
 - Confirm (do nothing, auto-approves in 4 hours)
 - Adjust quantity
 - Cancel this one time
6. Order auto-placed 

Safety limits:

- Max spending: 10M VND/week (customizable)
- Max 3 auto-orders/week
- Can pause anytime

Feature 18: Smart Notifications (Zero Spam)

Rules:

- Max 2 notifications/day (unless urgent)
- Send when shop owner is active (ML-predicted)
- Every notification = actionable (1-tap response)
- Easy opt-out/customize

Examples:

-  "Your rice is low - reorder now?" (actionable)
-  "Check out our new products!" (spam)

Feature 19: Delivery Preferences

- Set preferred delivery times (e.g., mornings only)
- AI schedules orders to arrive when convenient
- "Deliver Tuesday-Thursday between 9am-12pm"

Feature 20: Spending Insights

- "This month: 15M VND (↓10% vs last month)"
- "Top products: Rice (30%), Oil (20%), Noodles (15%)"
- "You saved 2M VND by using Aladin vs old supplier"

Tech Stack

- **Auto-order engine:** Python (cron jobs + ML predictions)
- **Notification system:** Firebase Cloud Messaging (FCM)
- **User behavior tracking:** Mixpanel or Amplitude

AI/ML Work

Model 3: Notification Timing

- Input: User activity patterns (when they open app)
- Output: Best time to send notification
- Algorithm: Simple clustering (morning/afternoon/evening users)

Model 4: Inventory Simulation

- Input: Order history, delivery time, sales velocity
- Output: Optimal reorder timing
- Algorithm: Basic reinforcement learning (Q-learning)

Team (Month 5)

- Add: 1 Mobile Developer (build native app for Month 6)
- Existing: Karthick + 2 backend + 1 ML engineer + designer

Total: 6 people

Success Metrics

-  30% of shops enable auto-reorder
-  95% satisfaction with auto-orders (no complaints)

-  60% notification click-through rate (vs 5-10% industry avg)
-  70% of orders via Zalo/app (fully digital)

Budget: \$35-40K

- Team: \$30K (add mobile dev)
 - Infrastructure: \$3K
 - Notification costs: \$2K (SMS fallback for critical alerts)
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MONTH 6 (APRIL 2026): GO LIVE

Theme: "Full Mobile App Launch"

Goal

Public launch with 500 shops using the app

What to Build

Feature 21: Native Mobile App (iOS + Android)

Why native app (vs just Zalo)?

- Better performance (faster, smoother)
- Richer features (camera, push notifications)
- Professional brand presence (app store)
- Easier future updates

But keep Zalo integration (many users prefer it)

Feature 22: Onboarding Flow

New user experience:

1. Download app / Open Zalo OA
2. Enter phone number → OTP verification
3. AI asks: "What products do you usually sell?"
4. AI suggests starter products based on shop type
5. "Place your first order - 5% discount!"
6. Tutorial: Voice, photo, or chat ordering

Goal: First order within 5 minutes of signup

Feature 23: Business Dashboard

Shop owner sees:

- Revenue this month vs last month
- Top selling products
- Inventory status (what's running low)
- Order history (filterable, exportable)
- Delivery schedule (upcoming orders)

Value: "Aladin helps me run my business better"

Feature 24: Referral Program

Incentive to grow:

- Refer another shop owner → Both get 100K VND credit
- Tracked via referral code
- Gamification: "3 more referrals to unlock Gold tier"

Goal: 20% of new customers from referrals

Feature 25: Polish & Performance

- App loads in <2 seconds
- Works on 3G (rural-optimized)
- Offline mode: Queue orders, sync later
- Dark mode (battery saving)
- Multiple languages: Vietnamese (Saigon dialect primary)

Launch Strategy

Week 1-2: Private Beta

- Invite 100 loyal customers
- Intensive user testing
- Fix critical bugs

Week 3: Soft Launch

- Open to all 200+ existing customers
- Monitor performance (crashes, errors)
- Gather feedback

Week 4: Public Launch

- Press release: "Aladin AI - Voice ordering for rural shops"
- Social media campaign
- Broker network pushes app adoption
- Target: 500 shops using app by end of month

Tech Stack

- **Mobile:** React Native (cross-platform)
- **Backend:** Node.js + PostgreSQL (same as before)
- **Analytics:** Mixpanel (user behavior tracking)
- **Crash reporting:** Sentry (bug tracking)
- **Push notifications:** Firebase Cloud Messaging

Team (Month 6)

- Add: 1 QA Engineer (testing, bug fixing)
- Add: 1 DevOps Engineer (scaling, deployment)
- Existing: Karthick + 2 backend + 2 mobile + 1 ML + designer

Total: 8 people

Success Metrics (Launch Month)

-  500 shops using app (50% of customer base)
-  70% of orders via app (vs 30% phone)
-  4.5+ star rating (App Store / Google Play)
-  <1% crash rate
-  NPS (Net Promoter Score): 50+ (excellent)

Budget: \$40-50K

- Team: \$35K (full team for launch push)
- Infrastructure: \$5K (scaling for 500+ users)
- Marketing: \$5K (launch campaign, influencers)

- App Store fees: \$200 (Apple + Google)

TOTAL 6-MONTH INVESTMENT

Month	Focus	Budget	Team Size
1	Zalo Chatbot	\$15-20K	3-4
2	Voice Input	\$20-25K	4-5
3	OCR (Photo)	\$25-30K	5
4	AI Intelligence	\$30-35K	5
5	Automation	\$35-40K	6
6	GO LIVE	\$40-50K	8

Total: \$165-200K

Recommended: Raise \$200K seed round in Month 1 to fund this

FEATURE PRIORITY (Must-Have vs Nice-to-Have)

MUST-HAVE (Launch Blockers)

1. Zalo chatbot (menu-based ordering)
2. Voice ordering (Vietnamese speech-to-text)
3. Photo ordering (OCR invoice)
4. Demand forecasting (predict stockouts)
5. Auto-reorder system
6. Mobile app (iOS + Android)
7. Order tracking & notifications
8. Business dashboard (revenue, top products)

NICE-TO-HAVE (Post-Launch)

1. Multi-shop management

2. Price intelligence (competitor monitoring)
3. Advanced analytics (profit margins, trends)
4. Loyalty program (points, rewards)
5. In-app chat support
6. Video tutorials

If timeline is tight, ship Must-Haves only. Add Nice-to-Haves in Month 7-9.

RISK MITIGATION

Risk 1: Can't Build Everything in 6 Months

Probability: HIGH

Mitigation:

- Week 1: Lock feature scope (NO changes after)
 - If falling behind: Cut nice-to-haves, keep must-haves
 - Option: Push launch to Month 7 (still acceptable)
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Risk 2: Voice/OCR Accuracy Too Low

Probability: MEDIUM

Mitigation:

- Set minimum viable accuracy: 80% (not 95%)
 - Human-in-the-loop: Fallback to manual for failures
 - Improve post-launch (80% → 90% → 95% over time)
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Risk 3: Users Don't Adopt App

Probability: MEDIUM

Mitigation:

- Month 1: Start with Zalo (users already have it)

- Gradual migration: Zalo → Native app
 - Incentives: "First order via app = 5% discount"
 - Personal onboarding: Be (COO) calls each customer
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Risk 4: Can't Hire Fast Enough

Probability: MEDIUM

Mitigation:

- Start recruiting NOW (Month 0)
 - Use contractors for short-term (ML engineer, designer)
 - Offshore option: Hire Vietnam-based contractors (cheaper, fast)
 - Karthick codes critical features himself if needed
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Risk 5: Budget Overruns

Probability: MEDIUM

Mitigation:

- Raise \$200-250K (buffer for overruns)
 - Track spending weekly (not monthly)
 - If tight: Use cheaper alternatives (self-host vs cloud APIs)
 - Worst case: Extend timeline to Month 7-8
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SUCCESS METRICS (6-Month Goals)

App Adoption

-  500 shops using app (50% of customer base)
-  70% of orders via app (vs phone)
-  4.5+ star rating (app stores)

AI Performance

-  80% voice recognition accuracy
-  85% OCR accuracy
-  70% demand forecast accuracy
-  30% shops on auto-reorder

Business Impact

-  AOV increase: +15% (recommendations)
-  Order frequency: +20% (easier to order)
-  Customer satisfaction: 90%+ (NPS: 50+)
-  Retention: 88%+ (app creates lock-in)

Efficiency Gains

-  Ordering time: 10 minutes → 10 seconds (60x faster)
 -  Shop owner time saved: 2 hours/day → 10 minutes/day
 -  Aladin ops: Handle 2x more orders with same team (automation)
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POST-LAUNCH ROADMAP (Month 7-12)

Once app is live, focus on:

Month 7-8: Optimization

- Improve AI accuracy (voice, OCR, forecasting)
- Fix bugs, optimize performance
- Scale infrastructure (1,000 users)

Month 9-10: Advanced Features

- Multi-shop management
- Price intelligence
- Advanced analytics dashboard

Month 11-12: Geographic Expansion

- Clone app to Dong Nai province
- Localize for regional dialects (Central, Northern Vietnam)
- Prep for Series A fundraising

IMMEDIATE ACTION PLAN (THIS MONTH)

Week 1 (Nov 13-19): Team & Budget

- [] **Quyet:** Secure \$200K seed funding (or commit to bootstrap)
- [] **Karthick:** Start recruiting (2 devs + 1 ML engineer)
- [] **Be:** Select 20 beta testers (high-trust customers)

Week 2 (Nov 20-26): Kickoff

- [] **Team:** Product kickoff meeting (lock scope, timeline)
- [] **Karthick:** Set up development environment (AWS, Zalo OA)
- [] **Designer:** Start UI/UX mockups (Zalo chatbot screens)

Week 3 (Nov 27-Dec 3): Start Building

- [] **Backend Dev:** Build API (products, orders, customers)
- [] **Mobile Dev:** Zalo Mini App integration
- [] **ML Engineer:** Collect voice samples (Vietnamese)

Week 4 (Dec 4-10): First Demo

- [] **Team:** Internal demo (chatbot MVP)
- [] **Be:** Test with 5 beta customers
- [] **All:** Gather feedback, iterate

TECHNOLOGY CHOICES (Build vs Buy)

Build In-House

- Zalo chatbot (unique to Vietnam)
- Order management system (core business logic)
- Product recommendations (proprietary data)
- Mobile app (brand differentiation)

Use APIs/SaaS

- Speech-to-Text: Zalo API (free) or Google Cloud (paid)
- OCR: PaddleOCR (open source) + Google Vision (backup)
- LLM: Claude API (for conversational AI, if needed)
- Notifications: Firebase Cloud Messaging (standard)
- Analytics: Mixpanel (easier than building)

Never Build (Too Complex) ❌

- Payment gateway (use ZaloPay, Momo)
 - SMS/Email (use Twilio, SendGrid)
 - Maps/Geolocation (use Google Maps API)
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COMPETITIVE LANDSCAPE (Vietnam B2B Commerce Apps)

Who's Building Similar?

- **Vinshop (Masan):** Corporate, slow, urban-focused
- **Telio:** B2B marketplace, not AI-focused
- **Tiki Ngon:** B2C, not B2B
- **NinjaMart (India):** Not in Vietnam yet

Why We'll Win

1. **AI-first:** Voice + OCR + Automation (they're catalog-first)
 2. **Rural-optimized:** Works on 3G, simple UI (they focus on urban)
 3. **Vietnamese-native:** Dialects, slang, local context (they use English AI)
 4. **Speed:** 6-month launch (they take 18-24 months)
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THE VISION (3 Years Out)

2026 (End of Year 1):

- 2,000 shops using Aladin AI
- 80% of orders via app
- \$100K monthly GMV via app

2027 (Year 2):

- 10,000 shops across 5 provinces
- AI driving 70% of ordering decisions
- Expand to Cambodia, Laos (similar markets)

2028 (Year 3):

- 50,000 shops across Southeast Asia
 - "AI-powered Shopify for traditional trade"
 - Series B (\$10-20M) to scale regionally
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FINAL CHECKLIST: ARE WE READY?

✅ Team Commitment

- ☐ All 3 founders full-time by Month 1
- ☐ Karthick ready to lead 8-person tech team
- ☐ Budget approved: \$200K for 6 months

✅ Customer Validation

- ☐ 20 beta testers committed
- ☐ Customers excited about voice/photo ordering
- ☐ Willingness to switch from phone to app: 80%+

✅ Technical Feasibility

- ☐ Zalo API access confirmed
- ☐ Vietnamese OCR tested (85%+ accuracy achievable)
- ☐ Voice recognition tested (80%+ accuracy achievable)

✅ Business Readiness

- ☐ Current operations stable (200 customers, profitable)
 - ☐ Inventory management working (can handle 2x orders)
 - ☐ Delivery capacity (can scale to 500 customers)
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LET'S SHIP THIS! 

